

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm. Examination

University Theory Examination October/November 2017

PH225 PHYSICAL PHARMACEUTICS

Date: 06/10/2017, Friday

Time: 1:30 p.m. to 4:30 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. The cgs units for poise are _____
[A] dyne sec/cm²
[B] dyne *sec*cm²
[C]dyne /sec*cm²
[D]dyne sec/cm
 2. Castor oil give the _____ type of flow
[A]Plastic Flow
[B]Pseudoplastic Flow
[C]Dilatant Flow
[D]Newtonian flow
 3. Capillary viscometer give the _____ viscosity
[A]Relative
[B]Absulate
[C]Both
[D] None
 4. In the cone and plate viscometer _____ is the non movment part of viscometer
[A] Plate
[B] Cone
[C] Bob
[D] Cup
 5. Surface tension is the force per unit length that must be applied _____ to the surface
[A] Parallar
[B] Vertical
[C] Horizontal
[D] Direct

6. Sparingly soluble means _____ parts of solvent required for 1 part of solute
[A] 30-100
[B] 300-500
[C] 500-1000
[D] 1000-10000
7. What is the iso tonicity of blood?
[A] 0.9%
[B] 9%
[C] 1%
[D] 90%
8. HLB value of w/o emulsifying agents is _____
[A] 1-2
[B] 3-8
[C] 9-10
[D] 12-15
9. In the capillary rise method, the diameter of capillary increase the height of liquid will _____
[A] Decrease
[B] Increase
[C] Not change
[D] None
10. tonicity of the solution can be measured by _____ method
[A] Colligative
[B] haemolitic
[C] both
[D] none
11. which bond form when ethanol dissolve in water?
[A] van der Waals
[B] Hydrogen
[C] covalent
[D] all
12. Olive oil has viscosity lower at _____ temperature.
[A] 0°C
[B] 25°C
[C] 50°C
[D] -50°C
13. At standard atmospheric condition what is the pressure?
[A] 760mm hg
[B] 1000mmhg
[C] 1500 mmhg
[D] 500 mmhg

14. When the transition takes place metastable to a stable form is known as _____.
- [A] Monotropic
[B] Reversible
[C] enantiotropic
[D] None
15. As per BCS classification class I drug have _____ solubility
- [A] Low
[B] poor
[C] high
[D] very poor
16. Polar solvent dissolve in _____ solvent
- [A] Polar
[B] Non Polar
[C] both
[D] none
17. Water is _____ type of solvent
- [A] Polar
[B] Semi Polar
[C] Non Polar
[D] All
18. Ions which are produced from ligands are
- [A] Cation
[B] Anion
[C] complex ion
[D] all
19. _____ can be used as complexing agent
- [A] cyclo dextrin
[B] poly ethylen glycol
[C] urea
[D] all
20. Theobroma oil has how many polymorph
- [A] I
[B] II
[C] III
[D] V

SECTION – II

Q 2 Attempt any **TWO** of the following

- | | | |
|----------|--|-----------|
| A | Write a note on – Microscopy methods for particles size determination. | 05 |
| B | How the thixotropy can be measured? | 05 |
| C | Short Note: HLB | 05 |

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